SMARTATTEND

Solution Requirements

Functional and non-functional requirements for SmartAttend

Project documentation • Requirement Analysis

Prepared for an AI-assisted attendance management application

1. Problem statement

Manual classroom attendance can consume teaching time and can be difficult to review consistently. SmartAttend aims to support an administrator with an AI-assisted workflow for identifying attendance from supported visual input and reviewing the resulting records.

2. Objectives

- Reduce repetitive manual attendance work.
- Provide a clear interface for administrators.
- Process supported visual input and present understandable results.
- Keep attendance results reviewable and, where implemented, retrievable.

3. Functional requirements

ID	Requirement	Priority	Acceptance criteria
FR-01	The system shall provide an administrator login or access control where authentication is configured.	High	Unauthorized users cannot access protected admin actions.
FR-02	The system shall provide a dashboard to access the attendance workflow.	High	Admin can identify the next action from the main screen.
FR-03	The system shall accept only supported input formats and enforce configured upload limits.	High	Unsupported or oversized files produce a clear message.
FR-04	The system shall analyze the supplied visual input using the recognition pipeline implemented by the project.	High	Processing returns results or a clear failure/warning.
FR-05	The system shall display attendance-related results for administrator review.	High	Results are visible with status/identity fields supported by the implementation.
FR-06	The system should allow attendance records to be saved, searched, or exported if these functions are in scope.	Medium	A completed session can be retrieved/exported when the feature is implemented.
FR-07	The system should provide useful error handling for invalid inputs and processing failures.	Medium	Errors explain the issue and a reasonable next step.

4. Non-functional requirements

ID	Category	Requirement
NFR-01	Usability	Dashboard controls and errors should be understandable to a non-technical administrator.
NFR-02	Performance	Processing time should be measured on representative classroom media; no fixed latency is assumed until tested.
NFR-03	Security	Credentials and student data must not be exposed in source control or public logs.
NFR-04	Privacy	Collect and retain only data needed for attendance; follow applicable consent and institutional policy.
NFR-05	Reliability	Invalid or corrupted media should fail gracefully without crashing the entire app.
NFR-06	Maintainability	Dependencies, setup steps and configuration should be documented.
NFR-07	Portability	The project should run in a documented Python environment with compatible dependencies.

5. Constraints and dependencies

• Results depend on camera quality, lighting, face visibility and the quality of any enrolled reference data.
• The project requires compatible Python packages and any model files or environment variables specified by the repository.
• Features such as database persistence, reports, webcam capture, liveness checks or engagement scoring must be verified against the code and are not assumed to exist solely from the project name.

6. Acceptance summary

The solution is acceptable for a prototype when an administrator can access the dashboard, provide supported input, receive attendance-related output, understand errors, and review results. Persistence and reporting should be evaluated separately if included in the final scope.